

PROJECT DEVELOPMENT PHASE

Smart Leave Management System

Development Environment

Platform Used ServiceNow Developer Instance

Development Modules

- ServiceNow App Studio
- Flow Designer
- Business Rules
- Client Scripts
- Custom Tables
- Dashboards and Reports
- Form Designer

Application Development

Step 1: Creation of Application

A custom application named:

Smart Leave Management System was created using ServiceNow App Studio.

Purpose

The application serves as a centralized platform for managing employee leave requests, approvals, validations, and reporting.

Application Components

- Custom Table
- Leave Request Form
- Approval Flow
- Business Rule
- Client Script
- Dashboard

- Reports

4. Custom Table Development

A custom table was created to store employee leave request information.

Table Name

Leave Request

Fields Created

Field Name | Description

Employee Name | Employee requesting leave

Leave Type | Type of leave requested

Start Date | Leave start date

End Date | Leave end date

Total Days | Total leave duration

Reason | Purpose of leave request

Approval Status | Approval state

Purpose

The table stores all leave request records and acts as the primary data repository for the application.

5. Form Design Development

A user-friendly form was designed for employee leave request submission.

Form Fields

Employee Information

- Employee Name

Leave Information

- Leave Type

- Start Date

- End Date

- Total Days

- Reason

Validation

Mandatory fields were configured to ensure complete leave request submission.

Benefits

- Improved usability

- Reduced data entry errors

- Standardized leave request collection

6. Leave Request Form Development

A custom leave request form was created to allow employees to submit leave requests.

Form Components

Field | Type

Employee Name | Single Line Text

Leave Type | Choice

Start Date | Date

End Date | Date

Total Days | Integer

Reason | Multi-Line Text

Approval Status | Choice

Functionality

Employees can submit leave requests using a structured and user-friendly form.

7. Workflow Development Using Flow Designer

A workflow was created using Flow Designer to automate the leave request lifecycle.

Flow Name

Leave Approval Flow

Trigger

Leave Request Created

The workflow starts automatically whenever a new leave request record is created.

Initial Action

Update Status = Pending Approval

Purpose

The leave request is automatically marked as pending approval.

Approval Action

An approval step was configured using:

Ask For Approval

Purpose

The leave request is routed to the reporting manager for review and approval.

Approval Rule

Anyone Approves

Conditional Logic

An IF condition was configured.

Condition

Approval State = Approved

Purpose

The workflow proceeds only when approval is granted.

Status Update

If Approved

- Approval Status = Approved

If Rejected

- Approval Status = Rejected

Purpose

Automatically updates leave request status based on manager decision.

8. Business Rule Development

A Business Rule was configured to automatically calculate the total number of leave days.

Business Rule Name

Calculate Total Leave Days

Purpose

Automatically calculates leave duration based on the selected start date and end date.

Example

Start Date : 23-Jun-2026

End Date : 25-Jun-2026

Total Days : 3

Benefits

- Eliminates manual calculations
- Improves accuracy
- Reduces processing time

9. Client Script Development

A Client Script was configured to validate leave dates before form submission.

Client Script Name

Leave Date Validation

Validation Rule

End Date cannot be earlier than Start Date.

Error Message

"End Date cannot be earlier than Start Date"

Purpose

Prevents invalid leave requests and improves data quality.

10. Dashboard Development

A dashboard was created to monitor leave requests and approval status.

Dashboard Components

- Total Requests
- Approved Requests
- Rejected Requests
- Pending Requests

Purpose

Provides real-time visibility into leave request activities and approval trends.

11. Reporting Features

Reports were created to analyze leave request data.

Report Types

- Total Leave Requests Report
- Approved Requests Report
- Rejected Requests Report
- Pending Requests Report

Benefits

- Improved monitoring
- Better decision making
- Enhanced reporting capabilities

Conclusion

The Project Development Phase successfully implemented the Smart Leave Management System using ServiceNow. The application includes custom tables, forms, Flow Designer approvals, Business Rules for automatic leave calculation, Client Scripts for validation, and dashboards for reporting. These features provide an efficient, automated, and user-friendly leave management solution.